TECHNICAL INFORMATION
for WG1605

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for WG1605

CONTENTS

[1] BRAKE HORSE POWER 2
[2] FUEL CONSUMPTION 4
[3] NOISE LEVEL 6
[4] AIR REQUIREMENTS 8
(1) Combustion Air Requirements (at 25 deg. C and 100 kPa) 8
(2) Cooling Air Requirements (at 25 deg. C and 100 kPa) 10
(3) Combustion and Cooling Air Requirements (at 25 deg. C and 100 kPa) 12
[5] EXHAUST GAS VOLUME 14
[6] HEAT REJECTION TO COOLANT 16
[7] WATER FLOW RATE 19
[8] CENTER OF GRAVITY 20
[9] MASS ELASTIC SYSTEMS 21
(1) Equivalent Torsional Vibration Data 21
[10] UNBALANCED FORCES 22
(1) Base Data 22
(2) Unbalanced Inertia Force and Couple 22
(3) An Example of Calculation 22
REVISION HISTORY 23

NOTES

This document is applied for Kubota WG1605 model for OEM.

This document is intend to provide installation guide for the engine to the application.

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[1] BRAKE HORSE POWER

SAE J1349

Model	Fuel	Output		Engine Speed (rpm)				
				2000	2400	2800	3200	3600
WG1605-G-E3	Gasoline	Net intermittent	kW	23.2	27.9	31.7	35.5	38.5
			HP	31.1	37.4	42.5	47.6	51.6
			PS	31.5	37.9	43.0	48.2	52.3
		Net continuous	kW	18.5	22.3	25.3	28.4	30.8
			HP	24.9	29.9	34.0	38.1	41.3
			PS	25.2	30.3	34.4	38.6	41.8

Model	Fuel	Output		Engine Speed (rpm)				
				2000	2400	2800	3200	3600
WG1605-L-E3	LPG	Net intermittent	kW	22.8	27.2	30.9	34.2	37.1
			HP	30.6	36.4	41.5	45.9	49.8
			PS	31.0	36.9	42.0	46.6	50.4
		Net continuous	kW	18.3	21.7	24.7	27.4	29.7
			HP	24.5	29.1	33.2	36.7	39.8
			PS	24.8	29.5	33.6	37.2	40.4

Model	Fuel	Output		Engine Speed (rpm)				
				2000	2400	2800	3200	3600
WG1605-GL-E3	Gasoline	Net intermittent	kW	23.2	27.9	31.7	35.5	38.5
			HP	31.1	37.4	42.5	47.6	51.6
			PS	31.5	37.9	43.0	48.2	52.3
		Net continuous	kW	18.5	22.3	25.3	28.4	30.8
			HP	24.9	29.9	34.0	38.1	41.3
			PS	25.2	30.3	34.4	38.6	41.8
	LPG	Net intermittent	kW	22.8	27.2	30.9	34.2	37.1
			HP	30.6	36.4	41.5	45.9	49.8
			PS	31.0	36.9	42.0	46.6	50.4
		Net continuous	kW	18.3	21.7	24.7	27.4	29.7
			HP	24.5	29.1	33.2	36.7	39.8
			PS	24.8	29.5	33.6	37.2	40.4

Model	Fuel	Output		Engine Speed (rpm)				
				2000	2400	2800	3200	3600
WG1605-N-E3	*Natural Gas	Net intermittent	kW	21.4	25.5	29.1	32.2	34.8
			HP	28.7	34.2	39.1	43.2	46.7
			PS	29.1	34.7	39.6	43.8	47.3
		Net continuous	kW	17.1	20.4	23.3	25.8	27.8
			HP	23.0	27.4	31.3	34.6	37.3
			PS	23.3	27.8	31.7	35.1	37.8

SAE J1349

Model	Fuel	Output		Engine Speed (rpm)				
				2000	2400	2800	3200	3600
WG1605-LN-E3	LPG	Net intermittent	kW	22.8	27.2	30.9	34.2	37.1
			HP	30.6	36.4	41.5	45.9	49.8
			PS	31.0	36.9	42.0	46.6	50.4
		Net continuous	kW	18.3	21.7	24.7	27.4	29.7
			HP	24.5	29.1	33.2	36.7	39.8
			PS	24.8	29.5	33.6	37.2	40.4
	*Natural Gas	Net intermittent	kW	21.4	25.5	29.1	32.2	34.8
			HP	28.7	34.2	39.1	43.2	46.7
			PS	29.1	34.7	39.6	43.8	47.3
		Net continuous	kW	17.1	20.4	23.3	25.8	27.8
			HP	23.0	27.4	31.3	34.6	37.3
			PS	23.3	27.8	31.7	35.1	37.8

Model	Fuel	Output		Engine Speed (rpm)				
				2000	2400	2800	3200	3600
WG1605-GLN-E3	Gasoline	Net intermittent	kW	23.2	27.9	31.7	35.5	38.5
			HP	31.1	37.4	42.5	47.6	51.6
			PS	31.5	37.9	43.0	48.2	52.3
		Net continuous	kW	18.5	22.3	25.3	28.4	30.8
			HP	24.9	29.9	34.0	38.1	41.3
			PS	25.2	30.3	34.4	38.6	41.8
	LPG	Net intermittent	kW	22.8	27.2	30.9	34.2	37.1
			HP	30.6	36.4	41.5	45.9	49.8
			PS	31.0	36.9	42.0	46.6	50.4
		Net continuous	kW	18.3	21.7	24.7	27.4	29.7
			HP	24.5	29.1	33.2	36.7	39.8
			PS	24.8	29.5	33.6	37.2	40.4
	*Natural Gas	Net intermittent	kW	21.4	25.5	29.1	32.2	34.8
			HP	28.7	34.2	39.1	43.2	46.7
			PS	29.1	34.7	39.6	43.8	47.3
		Net continuous	kW	17.1	20.4	23.3	25.8	27.8
			HP	23.0	27.4	31.3	34.6	37.3
			PS	23.3	27.8	31.7	35.1	37.8

■ NOTE

- **Conversion rates**
  - **1 kW = 1.35962 PS = 1.34048 HP**
  - **1 PS = 0.7355 kW = 0.985925 HP**
  - **1 HP = 0.7457 kW = 1.01428 PS**
- *** Fuel detail**

  Japanese standard gas
  - **lower heating value: 40.6 MJ/m^3 (9699 kcal/m^3, 1090 BTU/ft^3)**
  - **supply pressure: 0.98 to 2.45 kPa (2.27 to 5.68 oz/in^2)**

[2] FUEL CONSUMPTION

Specific at net intermittent (SAE J1349)

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -G-E3	Gasoline	Brake horse power	kW	7.7	13.3	18.4	23.2	27.9	31.7	35.5	38.5
			HP	10.3	17.8	24.6	31.1	37.4	42.5	47.6	51.6
			PS	10.4	18.0	25.0	31.5	37.9	43.0	48.2	52.3
		Fuel consumption	g/kWh	289.2	264.5	258.0	256.5	253.1	256.3	262.3	269.5
			g/HPh	215.7	197.2	192.4	191.3	188.7	191.1	195.6	201.0
			g/PSh	212.7	194.5	189.8	188.7	186.2	188.5	192.9	198.2
			lb/HPh	0.475	0.435	0.424	0.422	0.416	0.421	0.431	0.443

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -L-E3	LPG	Brake horse power	kW	8.1	13.4	18.2	22.8	27.2	30.9	34.2	37.1
			HP	10.9	17.9	24.4	30.6	36.4	41.5	45.9	49.8
			PS	11.1	18.2	24.7	31.0	36.9	42.0	46.6	50.4
		Fuel consumption	g/kWh	245.0	234.3	235.2	236.7	238.9	241.3	249.6	260.9
			g/HPh	182.7	174.7	175.4	176.5	178.1	179.9	186.1	194.6
			g/PSh	180.2	172.3	173.0	174.1	175.7	177.5	183.6	191.9
			lb/HPh	0.403	0.385	0.387	0.389	0.393	0.397	0.410	0.429

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -GL -E3	Gasoline	Brake horse power	kW	7.7	13.3	18.4	23.2	27.9	31.7	35.5	38.5
			HP	10.3	17.8	24.6	31.1	37.4	42.5	47.6	51.6
			PS	10.4	18.0	25.0	31.5	37.9	43.0	48.2	52.3
		Fuel consumption	g/kWh	289.2	264.5	258.0	256.5	253.1	256.3	262.3	269.5
			g/HPh	215.7	197.2	192.4	191.3	188.7	191.1	195.6	201.0
			g/PSh	212.7	194.5	189.8	188.7	186.2	188.5	192.9	198.2
			lb/HPh	0.475	0.435	0.424	0.422	0.416	0.421	0.431	0.443
	LPG	Brake horse power	kW	8.1	13.4	18.2	22.8	27.2	30.9	34.2	37.1
			HP	10.9	17.9	24.4	30.6	36.4	41.5	45.9	49.8
			PS	11.1	18.2	24.7	31.0	36.9	42.0	46.6	50.4
		Fuel consumption	g/kWh	245.0	234.3	235.2	236.7	238.9	241.3	249.6	260.9
			g/HPh	182.7	174.7	175.4	176.5	178.1	179.9	186.1	194.6
			g/PSh	180.2	172.3	173.0	174.1	175.7	177.5	183.6	191.9
			lb/HPh	0.403	0.385	0.387	0.389	0.393	0.397	0.410	0.429

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605- N-E3	Natural Gas	Brake horse power	kW	7.5	12.5	17.2	21.4	25.5	29.1	32.2	34.8
			HP	10.1	16.8	23.0	28.7	34.2	39.1	43.2	46.7
			PS	10.2	17.0	23.3	29.1	34.7	39.6	43.8	47.3
		Fuel consumption	g/kWh	252.7	223.0	222.1	222.0	225.5	230.6	235.3	238.9
			g/HPh	188.5	166.3	165.7	165.6	168.2	172.0	175.5	178.2
			g/PSh	185.9	164.0	163.3	163.3	165.9	169.6	173.1	175.7
			lb/HPh	0.416	0.367	0.365	0.365	0.371	0.379	0.387	0.393

Specific at net intermittent (SAE J1349)

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -LN- E3	LPG	Brake horse power	kW	8.1	13.4	18.2	22.8	27.2	30.9	34.2	37.1
			HP	10.9	17.9	24.4	30.6	36.4	41.5	45.9	49.8
			PS	11.1	18.2	24.7	31.0	36.9	42.0	46.6	50.4
		Fuel consumption	g/kWh	245.0	234.3	235.2	236.7	238.9	241.3	249.6	260.9
			g/HPh	182.7	174.7	175.4	176.5	178.1	179.9	186.1	194.6
			g/PSh	180.2	172.3	173.0	174.1	175.7	177.5	183.6	191.9
			lb/HPh	0.403	0.385	0.387	0.389	0.393	0.397	0.410	0.429
	Natural Gas	Brake horse power	kW	7.5	12.5	17.2	21.4	25.5	29.1	32.2	34.8
			HP	10.1	16.8	23.0	28.7	34.2	39.1	43.2	46.7
			PS	10.2	17.0	23.3	29.1	34.7	39.6	43.8	47.3
		Fuel consumption	g/kWh	252.7	223.0	222.1	222.0	225.5	230.6	235.3	238.9
			g/HPh	188.5	166.3	165.7	165.6	168.2	172.0	175.5	178.2
			g/PSh	185.9	164.0	163.3	163.3	165.9	169.6	173.1	175.7
			lb/HPh	0.416	0.367	0.365	0.365	0.371	0.379	0.387	0.393

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -GLN- E3	Gasoline	Brake horse power	kW	7.7	13.3	18.4	23.2	27.9	31.7	35.5	38.5
			HP	10.3	17.8	24.6	31.1	37.4	42.5	47.6	51.6
			PS	10.4	18.0	25.0	31.5	37.9	43.0	48.2	52.3
		Fuel consumption	g/kWh	289.2	264.5	258.0	256.5	253.1	256.3	262.3	269.5
			g/HPh	215.7	197.2	192.4	191.3	188.7	191.1	195.6	201.0
			g/PSh	212.7	194.5	189.8	188.7	186.2	188.5	192.9	198.2
			lb/HPh	0.475	0.435	0.424	0.422	0.416	0.421	0.431	0.443
	LPG	Brake horse power	kW	8.1	13.4	18.2	22.8	27.2	30.9	34.2	37.1
			HP	10.9	17.9	24.4	30.6	36.4	41.5	45.9	49.8
			PS	11.1	18.2	24.7	31.0	36.9	42.0	46.6	50.4
		Fuel consumption	g/kWh	245.0	234.3	235.2	236.7	238.9	241.3	249.6	260.9
			g/HPh	182.7	174.7	175.4	176.5	178.1	179.9	186.1	194.6
			g/PSh	180.2	172.3	173.0	174.1	175.7	177.5	183.6	191.9
			lb/HPh	0.403	0.385	0.387	0.389	0.393	0.397	0.410	0.429
	Natural Gas	Brake horse power	kW	7.5	12.5	17.2	21.4	25.5	29.1	32.2	34.8
			HP	10.1	16.8	23.0	28.7	34.2	39.1	43.2	46.7
			PS	10.2	17.0	23.3	29.1	34.7	39.6	43.8	47.3
		Fuel consumption	g/kWh	252.7	223.0	222.1	222.0	225.5	230.6	235.3	238.9
			g/HPh	188.5	166.3	165.7	165.6	168.2	172.0	175.5	178.2
			g/PSh	185.9	164.0	163.3	163.3	165.9	169.6	173.1	175.7
			lb/HPh	0.416	0.367	0.365	0.365	0.371	0.379	0.387	0.393

■ NOTE

- **Conversion rates**
  - **1 kW = 1.35962 PS = 1.34048 HP**
  - **1 PS = 0.7355 kW = 0.985925 HP**
  - **1 HP = 0.7457 kW = 1.01428 PS**
  - **1 kg = 2.20462 lb (1 g = 0.00220462 lb)**
  - **1 lb = 0.45359 kg**

[3] NOISE LEVEL

unit: dB (A)

	rpm	Load ratio	Fuel	Sound pressure at 1.0 m (3.3 ft)
WG1605-G-E3	3600	0/4	Gasoline	91.5
	3600	4/4		93.5
	3000	0/4		87.5
	3000	4/4		89.4
	750	0/4		68.7

	rpm	Load ratio	Fuel	Sound pressure at 1.0 m (3.3 ft)
WG1605-L-E3	3600	0/4	LPG	91.0
	3600	4/4		92.3
	3000	0/4		87.6
	3000	4/4		88.8
	750	0/4		68.8

	rpm	Load ratio	Fuel	Sound pressure at 1.0 m (3.3 ft)
WG1605-GL-E3	3600	0/4	Gasoline	91.5
	3600	4/4		93.5
	3000	0/4		87.5
	3000	4/4		89.4
	750	0/4		68.7
	3600	0/4	LPG	91.0
	3600	4/4		92.3
	3000	0/4		87.6
	3000	4/4		88.8
	750	0/4		68.8

	rpm	Load ratio	Fuel	Sound pressure at 1.0 m (3.3 ft)
WG1605-N-E3	3600	0/4	Natural Gas	90.8
	3600	4/4		92.0
	3000	0/4		87.4
	3000	4/4		88.6
	750	0/4		68.8

	rpm	Load ratio	Fuel	Sound pressure at 1.0 m (3.3 ft)
WG1605-LN-E3	3600	0/4	LPG	91.0
	3600	4/4		92.3
	3000	0/4		87.6
	3000	4/4		88.8
	750	0/4		68.8
	3600	0/4	Natural Gas	90.8
	3600	4/4		92.0
	3000	0/4		87.4
	3000	4/4		88.6
	750	0/4		68.8

unit: dB (A)

	rpm	Load ratio	Fuel	Sound pressure at 1.0 m (3.3 ft)
WG1605-GLN -E3	3600	0/4	Gasoline	91.5
	3600	4/4		93.5
	3000	0/4		87.5
	3000	4/4		89.4
	750	0/4		68.7
	3600	0/4	LPG	91.0
	3600	4/4		92.3
	3000	0/4		87.6
	3000	4/4		88.8
	750	0/4		68.8
	3600	0/4	Natural Gas	90.8
	3600	4/4		92.0
	3000	0/4		87.4
	3000	4/4		88.6
	750	0/4		68.8

■ **NOTE**

- **Measurement conditions: with radiator, cooling fan, air cleaner and muffler.**

[4] AIR REQUIREMENTS

(1) Combustion Air Requirements (at 25 deg. C and 100 kPa)

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -G-E3	Gasoline	Combustion air requirements	L/sec	8.4	12.3	16.7	20.9	25.0	29.3	33.1	36.2
			m^3/h	30.3	44.1	60.1	75.1	90.0	105.3	119.1	130.2
			in^3/sec	513	748	1019	1273	1526	1785	2019	2207
			ft^3/min	17.8	26.0	35.4	44.2	53.0	62.0	70.1	76.6

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -L-E3	LPG	Combustion air requirements	L/sec	8.4	12.3	16.7	20.9	25.0	29.3	33.1	36.2
			m^3/h	30.3	44.1	60.1	75.1	90.0	105.3	119.1	130.2
			in^3/sec	513	748	1019	1273	1526	1785	2019	2207
			ft^3/min	17.8	26.0	35.4	44.2	53.0	62.0	70.1	76.6

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -GL -E3	Gasoline	Combustion air requirements	L/sec	8.4	12.3	16.7	20.9	25.0	29.3	33.1	36.2
			m^3/h	30.3	44.1	60.1	75.1	90.0	105.3	119.1	130.2
			in^3/sec	513	748	1019	1273	1526	1785	2019	2207
			ft^3/min	17.8	26.0	35.4	44.2	53.0	62.0	70.1	76.6
	LPG	Combustion air requirements	L/sec	8.4	12.3	16.7	20.9	25.0	29.3	33.1	36.2
			m^3/h	30.3	44.1	60.1	75.1	90.0	105.3	119.1	130.2
			in^3/sec	513	748	1019	1273	1526	1785	2019	2207
			ft^3/min	17.8	26.0	35.4	44.2	53.0	62.0	70.1	76.6

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -N-E3	Natural Gas	Combustion air requirements	L/sec	6.6	9.7	13.3	16.6	20.1	23.4	26.3	28.8
			m^3/h	23.8	34.8	48.0	59.8	72.2	84.3	94.8	103.7
			in^3/sec	404	590	814	1014	1224	1430	1607	1758
			ft^3/min	14.0	20.5	28.3	35.2	42.5	49.6	55.8	61.1

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -LN -E3	LPG	Combustion air requirements	L/sec	8.4	12.3	16.7	20.9	25.0	29.3	33.1	36.2
			m^3/h	30.3	44.1	60.1	75.1	90.0	105.3	119.1	130.2
			in^3/sec	513	748	1019	1273	1526	1785	2019	2207
			ft^3/min	17.8	26.0	35.4	44.2	53.0	62.0	70.1	76.6
	Natural Gas	Combustion air requirements	L/sec	6.6	9.7	13.3	16.6	20.1	23.4	26.3	28.8
			m^3/h	23.8	34.8	48.0	59.8	72.2	84.3	94.8	103.7
			in^3/sec	404	590	814	1014	1224	1430	1607	1758
			ft^3/min	14.0	20.5	28.3	35.2	42.5	49.6	55.8	61.1

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -GLN -E3	Gasoline	Combustion air requirements	L/sec	8.4	12.3	16.7	20.9	25.0	29.3	33.1	36.2
			m^3/h	30.3	44.1	60.1	75.1	90.0	105.3	119.1	130.2
			in^3/sec	513	748	1019	1273	1526	1785	2019	2207
			ft^3/min	17.8	26.0	35.4	44.2	53.0	62.0	70.1	76.6
	LPG	Combustion air requirements	L/sec	8.4	12.3	16.7	20.9	25.0	29.3	33.1	36.2
			m^3/h	30.3	44.1	60.1	75.1	90.0	105.3	119.1	130.2
			in^3/sec	513	748	1019	1273	1526	1785	2019	2207
			ft^3/min	17.8	26.0	35.4	44.2	53.0	62.0	70.1	76.6
	Natural Gas	Combustion air requirements	L/sec	6.6	9.7	13.3	16.6	20.1	23.4	26.3	28.8
			m^3/h	23.8	34.8	48.0	59.8	72.2	84.3	94.8	103.7
			in^3/sec	404	590	814	1014	1224	1430	1607	1758
			ft^3/min	14.0	20.5	28.3	35.2	42.5	49.6	55.8	61.1

■ **NOTE**

- **Combustion air requirements calculating formula**
- **$Q1 = Vh \times N \times C \times \eta \times 10^{-3}$**
  - **Q1: Amount of intake air (m^3/min)**
  - **Vh: Total displacement (L)**
  - **N: Engine speed (rpm)**
  - **C: Coefficient = 0.5**
  - **η: Intake efficiency**
    Gasoline: 0.85
    LPG: 0.81
    Natural Gas: 0.77

(2) Cooling Air Requirements (at 25 deg. C and 100 kPa)

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -G-E3	Gasoline	Cooling air requirements	L/sec	192.6	296.2	395.5	617.7	740.9	841.1	913.1	984.8
			m^3/h	693	1066	1424	2224	2667	3028	3287	3545
			in^3/sec	11753	18074	24137	37694	45210	51324	55720	60098
			ft^3/min	408.1	627.6	838.1	1308.8	1569.8	1782.1	1934.7	2086.7

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -L-E3	LPG	Cooling air requirements	L/sec	182.6	261.5	353.0	684.2	820.7	931.7	1011.5	1090.9
			m^3/h	657	941	1271	2463	2954	3354	3641	3927
			in^3/sec	11140	15959	21541	41755	50080	56853	61723	66572
			ft^3/min	386.8	554.1	748.0	1449.8	1738.9	1974.1	2143.2	2311.5

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -GL -E3	Gasoline	Cooling air requirements	L/sec	192.6	296.2	395.5	617.7	740.9	841.1	913.1	984.8
			m^3/h	693	1066	1424	2224	2667	3028	3287	3545
			in^3/sec	11753	18074	24137	37694	45210	51324	55720	60098
			ft^3/min	408.1	627.6	838.1	1308.8	1569.8	1782.1	1934.7	2086.7
	LPG	Cooling air requirements	L/sec	182.6	261.5	353.0	684.2	820.7	931.7	1011.5	1090.9
			m^3/h	657	941	1271	2463	2954	3354	3641	3927
			in^3/sec	11140	15959	21541	41755	50080	56853	61723	66572
			ft^3/min	386.8	554.1	748.0	1449.8	1738.9	1974.1	2143.2	2311.5

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -N-E3	Natural Gas	Cooling air requirements	L/sec	256	369	524	586	722	814	889	916
			m^3/h	923	1329	1888	2111	2598	2931	3199	3297
			in^3/sec	15651	22525	32005	35785	44034	49680	54232	55890
			ft^3/min	543.4	782.1	1111.3	1242.5	1529.0	1725.0	1883.1	1940.6

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -LN -E3	LPG	Cooling air requirements	L/sec	182.6	261.5	353.0	684.2	820.7	931.7	1011.5	1090.9
			m^3/h	657	941	1271	2463	2954	3354	3641	3927
			in^3/sec	11140	15959	21541	41755	50080	56853	61723	66572
			ft^3/min	386.8	554.1	748.0	1449.8	1738.9	1974.1	2143.2	2311.5
	Natural Gas	Cooling air requirements	L/sec	256	369	524	586	722	814	889	916
			m^3/h	923	1329	1888	2111	2598	2931	3199	3297
			in^3/sec	15651	22525	32005	35785	44034	49680	54232	55890
			ft^3/min	543	782	1111	1243	1529	1725	1883	1941

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -GLN -E3	Gasoline	Cooling air requirements	L/sec	192.6	296.2	395.5	617.7	740.9	841.1	913.1	984.8
			m^3/h	693	1066	1424	2224	2667	3028	3287	3545
			in^3/sec	11753	18074	24137	37694	45210	51324	55720	60098
			ft^3/min	408.1	627.6	838.1	1308.8	1569.8	1782.1	1934.7	2086.7
	LPG	Cooling air requirements	L/sec	182.6	261.5	353.0	684.2	820.7	931.7	1011.5	1090.9
			m^3/h	657	941	1271	2463	2954	3354	3641	3927
			in^3/sec	11140	15959	21541	41755	50080	56853	61723	66572
			ft^3/min	386.8	554.1	748.0	1449.8	1738.9	1974.1	2143.2	2311.5
	Natural Gas	Cooling air requirements	L/sec	256	369	524	586	722	814	889	916
			m^3/h	923	1329	1888	2111	2598	2931	3199	3297
			in^3/sec	15651	22525	32005	35785	44034	49680	54232	55890
			ft^3/min	543	782	1111	1243	1529	1725	1883	1941

■ NOTE

- **Above data is decided by following conditions.**
  - **Using the standard radiator.**
  - **Engine is run as open unit.**

(3) Combustion and Cooling Air Requirements (at 25 deg. C and 100 kPa)

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -G-E3	Gasoline	Combustion and cooling air requirements	L/sec	201	308	412	639	766	870	946	1021
			m^3/h	724	1110	1484	2299	2757	3133	3406	3676
			in^3/sec	12266	18822	25156	38967	46736	53109	57739	62304
			ft^3/min	426	654	873	1353	1623	1844	2005	2163

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -L-E3	LPG	Combustion and cooling air requirements	L/sec	191	274	370	705	846	961	1045	1127
			m^3/h	687	986	1331	2538	3044	3459	3760	4057
			in^3/sec	11653	16707	22561	43028	51606	58638	63742	68779
			ft^3/min	405	580	783	1494	1792	2036	2213	2388

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -GL -E3	Gasoline	Combustion and cooling air requirements	L/sec	201	308	412	639	766	870	946	1021
			m^3/h	724	1110	1484	2299	2757	3133	3406	3676
			in^3/sec	12266	18822	25156	38967	46736	53109	57739	62304
			ft^3/min	426	654	873	1353	1623	1844	2005	2163
	LPG	Combustion and cooling air requirements	L/sec	191	274	370	705	846	961	1045	1127
			m^3/h	687	986	1331	2538	3044	3459	3760	4057
			in^3/sec	11653	16707	22561	43028	51606	58638	63742	68779
			ft^3/min	405	580	783	1494	1792	2036	2213	2388

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -N-E3	Natural Gas	Combustion and cooling air requirements	L/sec	263	379	538	603	742	838	915	945
			m^3/h	947	1364	1936	2171	2670	3015	3294	3401
			in^3/sec	16055	23116	32819	36799	45258	51109	55839	57648
			ft^3/min	557	803	1140	1278	1571	1775	1939	2002

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -LN -E3	LPG	Combustion and cooling air requirements	L/sec	191	274	370	705	846	961	1045	1127
			m^3/h	687	986	1331	2538	3044	3459	3760	4057
			in^3/sec	11653	16707	22561	43028	51606	58638	63742	68779
			ft^3/min	405	580	783	1494	1792	2036	2213	2388
	Natural Gas	Combustion and cooling air requirements	L/sec	263	379	538	603	742	838	915	945
			m^3/h	947	1364	1936	2171	2670	3015	3294	3401
			in^3/sec	16055	23116	32819	36799	45258	51109	55839	57648
			ft^3/min	557	803	1140	1278	1571	1775	1939	2002

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -GLN -E3	Gasoline	Combustion and cooling air requirements	L/sec	201	308	412	639	766	870	946	1021
			m^3/h	724	1110	1484	2299	2757	3133	3406	3676
			in^3/sec	12266	18822	25156	38967	46736	53109	57739	62304
			ft^3/min	426	654	873	1353	1623	1844	2005	2163
	LPG	Combustion and cooling air requirements	L/sec	191	274	370	705	846	961	1045	1127
			m^3/h	687	986	1331	2538	3044	3459	3760	4057
			in^3/sec	11653	16707	22561	43028	51606	58638	63742	68779
			ft^3/min	405	580	783	1494	1792	2036	2213	2388
	Natural Gas	Combustion and cooling air requirements	L/sec	263	379	538	603	742	838	915	945
			m^3/h	947	1364	1936	2171	2670	3015	3294	3401
			in^3/sec	16055	23116	32819	36799	45258	51109	55839	57648
			ft^3/min	557	803	1140	1278	1571	1775	1939	2002

■ **NOTE**

- **Cooling fan and fan pulley specifications (Cooling fan Part No. 15442-7411Δ)**

Item	
Fan diameter	350 mm (13.78 in)
No. of blades and type of shape	5 blades and straight type
Diameter of fan driving pulley	105 mm (4.13 in)
Diameter of fan pulley	96 mm (3.78 in)

- **Conversion rates**

  1 L = 61.0237 in^3 = 0.035315 ft^3
  1 ft^3 = 28.3168 L
  1 L/sec = 3.6 m^3/h = 2.1189 ft^3/min

[5] EXHAUST GAS VOLUME

at 25 deg. C and 100 kPa

Model	Fuel	Output		Engine Speed (rpm)				
				2000	2400	2800	3200	3600
WG1605-G-E3	Gasoline	Gas volume	L/sec	68.4	81.0	93.2	102.8	109.0
			m^3/h	246.3	291.5	335.4	370.2	392.5
			in^3/sec	4174	4941	5685	6275	6653
			ft^3/min	144.9	171.6	197.4	217.9	231.0

Model	Fuel	Output		Engine Speed (rpm)				
				2000	2400	2800	3200	3600
WG1605-L-E3	LPG	Gas volume	L/sec	65.2	77.2	88.8	98.0	103.9
			m^3/h	234.7	277.8	319.6	352.8	374.0
			in^3/sec	3978	4709	5417	5980	6340
			ft^3/min	138.1	163.5	188.1	207.6	220.1

Model	Fuel	Output		Engine Speed (rpm)				
				2000	2400	2800	3200	3600
WG1605-GL-E3	Gasoline	Gas volume	L/sec	68.4	81.0	93.2	102.8	109.0
			m^3/h	246.3	291.5	335.4	370.2	392.5
			in^3/sec	4174	4941	5685	6275	6653
			ft^3/min	144.9	171.6	197.4	217.9	231.0
	LPG	Gas volume	L/sec	65.2	77.2	88.8	98.0	103.9
			m^3/h	234.7	277.8	319.6	352.8	374.0
			in^3/sec	3978	4709	5417	5980	6340
			ft^3/min	138.1	163.5	188.1	207.6	220.1

Model	Fuel	Output		Engine Speed (rpm)				
				2000	2400	2800	3200	3600
WG1605-N-E3	Natural Gas	Gas volume	L/sec	53.4	65.2	75.6	83.4	89.9
			m^3/h	192.3	234.9	272.3	300.1	323.7
			in^3/sec	3260	3982	4615	5087	5487
			ft^3/min	113.2	138.2	160.2	176.7	190.5

Model	Fuel	Output		Engine Speed (rpm)				
				2000	2400	2800	3200	3600
WG1605-LN-E3	LPG	Gas volume	L/sec	65.2	77.2	88.8	98.0	103.9
			m^3/h	234.7	277.8	319.6	352.8	374.0
			in^3/sec	3978	4709	5417	5980	6340
			ft^3/min	138.1	163.5	188.1	207.6	220.1
	Natural Gas	Gas volume	L/sec	53.4	65.2	75.6	83.4	89.9
			m^3/h	192.3	234.9	272.3	300.1	323.7
			in^3/sec	3260	3982	4615	5087	5487
			ft^3/min	113.2	138.2	160.2	176.7	190.5

at 25 deg. C and 100 kPa

Model	Fuel	Output		Engine Speed (rpm)				
				2000	2400	2800	3200	3600
WG1605 -GLN-E3	Gasoline	Gas volume	L/sec	68.4	81.0	93.2	102.8	109
			m^3/h	246.3	291.5	335.4	370.2	392.5
			in^3/sec	4174	4941	5685	6275	6653
			ft^3/min	144.9	171.6	197.4	217.9	231.0
	LPG	Gas volume	L/sec	65.2	77.2	88.8	98.0	103.9
			m^3/h	234.7	277.8	319.6	352.8	374.0
			in^3/sec	3978	4709	5417	5980	6340
			ft^3/min	138.1	163.5	188.1	207.6	220.1
	Natural Gas	Gas volume	L/sec	53.4	65.2	75.6	83.4	89.9
			m^3/h	192.3	234.9	272.3	300.1	323.7
			in^3/sec	3260	3982	4615	5087	5487
			ft^3/min	113.2	138.2	160.2	176.7	190.5

■ NOTE

- **Conversion rates**
  1 L = 61.0237 in^3 = 0.035315 ft^3
  1 ft^3 = 28.3168 L
  1 L/sec = 3.6 m^3/h = 127.133 ft^3/min

[6] HEAT REJECTION TO COOLANT

Specific at net intermittent (SAE J1349)

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -G-E3	Gasoline	Brake horse power	kW	7.7	13.3	18.4	23.2	27.9	31.7	35.5	38.5
			HP	10.3	17.8	24.6	31.1	37.4	42.5	47.6	51.6
			PS	10.4	18.0	25.0	31.5	37.9	43.0	48.2	52.3
		Fuel consumption	g/kWh	289.2	264.5	258.0	256.5	253.1	256.3	262.3	269.5
			g/HPh	215.7	197.2	192.4	191.3	188.7	191.1	195.6	201.0
			g/PSh	212.7	194.5	189.8	188.7	186.2	188.5	192.9	198.2
			lb/HPh	0.475	0.435	0.424	0.422	0.416	0.421	0.431	0.443
		Heat rejection to cooling water	MJ/h	20.39	31.36	41.88	65.40	78.44	89.05	96.68	104.28
			kcal/h	4872	7492	10005	15624	18739	21274	23096	24910
			BTU/h	19330	29727	39698	61996	74357	84414	91644	98843

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -L-E3	LPG	Brake horse power	kW	8.1	13.4	18.2	22.8	27.2	30.9	34.2	37.1
			HP	10.9	17.9	24.4	30.6	36.4	41.5	45.9	49.8
			PS	11.1	18.2	24.7	31.0	36.9	42.0	46.6	50.4
		Fuel consumption	g/kWh	245.0	234.3	235.2	236.7	238.9	241.3	249.6	260.9
			g/HPh	182.7	174.7	175.4	176.5	178.1	179.9	186.1	194.6
			g/PSh	180.2	172.3	173.0	174.1	175.7	177.5	183.6	191.9
			lb/HPh	0.403	0.385	0.387	0.389	0.393	0.397	0.41	0.429
		Heat rejection to cooling water	MJ/h	19.33	27.69	37.38	72.45	86.89	98.65	107.10	115.51
			kcal/h	4618	6615	8929	17307	20758	23565	25584	27594
			BTU/h	18322	26248	35430	68674	82368	93507	101517	109492

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -GL -E3	Gasoline	Brake horse power	kW	7.7	13.3	18.4	23.2	27.9	31.7	35.5	38.5
			HP	10.3	17.8	24.6	31.1	37.4	42.5	47.6	51.6
			PS	10.4	18.0	25.0	31.5	37.9	43.0	48.2	52.3
		Fuel consumption	g/kWh	289.2	264.5	258.0	256.5	253.1	256.3	262.3	269.5
			g/HPh	215.7	197.2	192.4	191.3	188.7	191.1	195.6	201.0
			g/PSh	212.7	194.5	189.8	188.7	186.2	188.5	192.9	198.2
			lb/HPh	0.475	0.435	0.424	0.422	0.416	0.421	0.431	0.443
		Heat rejection to cooling water	MJ/h	20.39	31.36	41.88	65.40	78.44	89.05	96.68	104.28
			kcal/h	4872	7492	10005	15624	18739	21274	23096	24910
			BTU/h	19330	29727	39698	61996	74357	84414	91644	98843
	LPG	Brake horse power	kW	8.1	13.4	18.2	22.8	27.2	30.9	34.2	37.1
			HP	10.9	17.9	24.4	30.6	36.4	41.5	45.9	49.8
			PS	11.1	18.2	24.7	31.0	36.9	42.0	46.6	50.4
		Fuel consumption	g/kWh	245.0	234.3	235.2	236.7	238.9	241.3	249.6	260.9
			g/HPh	182.7	174.7	175.4	176.5	178.1	179.9	186.1	194.6
			g/PSh	180.2	172.3	173.0	174.1	175.7	177.5	183.6	191.9
			lb/HPh	0.403	0.385	0.387	0.389	0.393	0.397	0.410	0.429
		Heat rejection to cooling water	MJ/h	19.33	27.69	37.38	72.45	86.89	98.65	107.10	115.51
			kcal/h	4618	6615	8929	17307	20758	23565	25584	27594
			BTU/h	18322	26248	35430	68674	82368	93507	101517	109492

Specific at net intermittent (SAE J1349)

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -N-E3	Natural Gas	Brake horse power	kW	7.5	12.5	17.2	21.4	25.5	29.1	32.2	34.8
			HP	10.1	16.8	23.0	28.7	34.2	39.1	43.2	46.7
			PS	10.2	17.0	23.3	29.1	34.7	39.6	43.8	47.3
		Fuel consumption	g/kWh	252.7	223.0	222.1	222.0	225.5	230.6	235.3	238.9
			g/HPh	188.5	166.3	165.7	165.6	168.2	172.0	175.5	178.2
			g/PSh	185.9	164.0	163.3	163.3	165.9	169.6	173.1	175.7
			lb/HPh	0.416	0.367	0.365	0.365	0.371	0.379	0.387	0.393
		Heat rejection to cooling water	MJ/h	27.2	39.1	55.5	62.1	76.4	86.2	94.1	97.0
			kcal/h	6487	9337	13266	14833	18252	20592	22479	23166
			BTU/h	25739	37044	52634	58850	72417	81701	89188	91913

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -LN -E3	LPG	Brake horse power	kW	8.1	13.4	18.2	22.8	27.2	30.9	34.2	37.1
			HP	10.9	17.9	24.4	30.6	36.4	41.5	45.9	49.8
			PS	11.1	18.2	24.7	31.0	36.9	42.0	46.6	50.4
		Fuel consumption	g/kWh	245.0	234.3	235.2	236.7	238.9	241.3	249.6	260.9
			g/HPh	182.7	174.7	175.4	176.5	178.1	179.9	186.1	194.6
			g/PSh	180.2	172.3	173.0	174.1	175.7	177.5	183.6	191.9
			lb/HPh	0.403	0.385	0.387	0.389	0.393	0.397	0.410	0.429
		Heat rejection to cooling water	MJ/h	19.33	27.69	37.38	72.45	86.89	98.65	107.10	115.51
			kcal/h	4618	6615	9829	17307	20758	23565	25584	27594
			BTU/h	18322	26248	35430	68674	82368	93507	101517	109492
	Natural Gas	Brake horse power	kW	7.5	12.5	17.2	21.4	25.5	29.1	32.2	34.8
			HP	10.1	16.8	23.0	28.7	34.2	39.1	43.2	46.7
			PS	10.2	17.0	23.3	29.1	34.7	39.6	43.8	47.3
		Fuel consumption	g/kWh	252.7	223.0	222.1	222.0	225.5	230.6	235.3	238.9
			g/HPh	188.5	166.3	165.7	165.6	168.2	172.0	175.5	178.2
			g/PSh	185.9	164.0	163.3	163.3	165.9	169.6	173.1	175.7
			lb/HPh	0.416	0.367	0.365	0.365	0.371	0.379	0.387	0.393
		Heat rejection to cooling water	MJ/h	27.2	39.1	55.5	62.1	76.4	86.2	94.1	97.0
			kcal/h	6487	9337	13266	14833	18252	20592	22479	23166
			BTU/h	25739	37044	52634	58850	72417	81701	89188	91913

Specific at net intermittent (SAE J1349)

Model	Fuel	Output		Engine Speed (rpm)							
				800	1200	1600	2000	2400	2800	3200	3600
WG 1605 -GLN -E3	Gasoline	Brake horse power	kW	7.7	13.3	18.4	23.2	27.9	31.7	35.5	38.5
			HP	10.3	17.8	24.6	31.1	37.4	42.5	47.6	51.6
			PS	10.4	18.0	25.0	31.5	37.9	43.0	48.2	52.3
		Fuel consumption	g/kWh	289.2	264.5	258.0	256.5	253.1	256.3	262.3	269.5
			g/HPh	215.7	197.2	192.4	191.3	188.7	191.1	195.6	201.0
			g/PSh	212.7	194.5	189.8	188.7	186.2	188.5	192.9	198.2
			lb/HPh	0.475	0.435	0.424	0.422	0.416	0.421	0.431	0.443
		Heat rejection to cooling water	MJ/h	20.39	31.36	41.88	65.40	78.44	89.05	96.68	104.28
			kcal/h	4872	7492	10005	15624	18739	21274	23096	24910
			BTU/h	19330	29727	39698	61996	74357	84414	91644	98843
	LPG	Brake horse power	kW	8.1	13.4	18.2	22.8	27.2	30.9	34.2	37.1
			HP	10.9	17.9	24.4	30.6	36.4	41.5	45.9	49.8
			PS	11.1	18.2	24.7	31.0	36.9	42.0	46.6	50.4
		Fuel consumption	g/kWh	245.0	234.3	235.2	236.7	238.9	241.3	249.6	260.9
			g/HPh	182.7	174.7	175.4	176.5	178.1	179.9	186.1	194.6
			g/PSh	180.2	172.3	173.0	174.1	175.7	177.5	183.6	191.9
			lb/HPh	0.403	0.385	0.387	0.389	0.393	0.397	0.410	0.429
		Heat rejection to cooling water	MJ/h	19.33	27.69	37.38	72.45	86.89	98.65	107.10	115.51
			kcal/h	4618	6615	9829	17307	20758	23565	25584	27594
			BTU/h	18322	26248	35430	68674	82368	93507	101517	109492
	Natural Gas	Brake horse power	kW	7.5	12.5	17.2	21.4	25.5	29.1	32.2	34.8
			HP	10.1	16.8	23.0	28.7	34.2	39.1	43.2	46.7
			PS	10.2	17.0	23.3	29.1	34.7	39.6	43.8	47.3
		Fuel consumption	g/kWh	252.7	223.0	222.1	222.0	225.5	230.6	235.3	238.9
			g/HPh	188.5	166.3	165.7	165.6	168.2	172.0	175.5	178.2
			g/PSh	185.9	164.0	163.3	163.3	165.9	169.6	173.1	175.7
			lb/HPh	0.416	0.367	0.365	0.365	0.371	0.379	0.387	0.393
		Heat rejection to cooling water	MJ/h	27.2	39.1	55.5	62.1	76.4	86.2	94.1	97.0
			kcal/h	6487	9337	13266	14833	18252	20592	22479	23166
			BTU/h	25739	37044	52634	58850	72417	81701	89188	91913

■ **NOTE**

- **Heat rejection to cooling water calculating formula**
- **Ho = Hu x Ne x be x i**

  Ho: Heat rejection to cooling water
  Hu: Fuel lower heating value
  Gasoline: 42.7 MJ/kg (10201 kcal/kg, 18361 BTU/lb)
  LPG: 47.3 MJ/kg (11300 kcal/kg, 20339 BTU/lb)
  Natural Gas: 49.5 MJ/kg (11828 kcal/kg, 21285 BTU/lb)
  Japanese standard gas
  lower heating value: 40.6 MJ/m^3 (9699 kcal/m^3, 1090 BTU/ft^3)
  supply pressure: 0.98 to 2.45 kPa (2.27 to 5.68 oz/in^2)
  Ne: Brake horse power
  be: Specific fuel consumption ratio
  i: Dispersion ratio to cooling water

[7] WATER FLOW RATE

Water Pump	16251-7303Δ, EG523-7303Δ
Fan Pulley Dia.	96 mm (3.78 in)
Fan Drive Pulley Dia.	105 mm (4.13 in)
Thermostat	19434-7301Δ

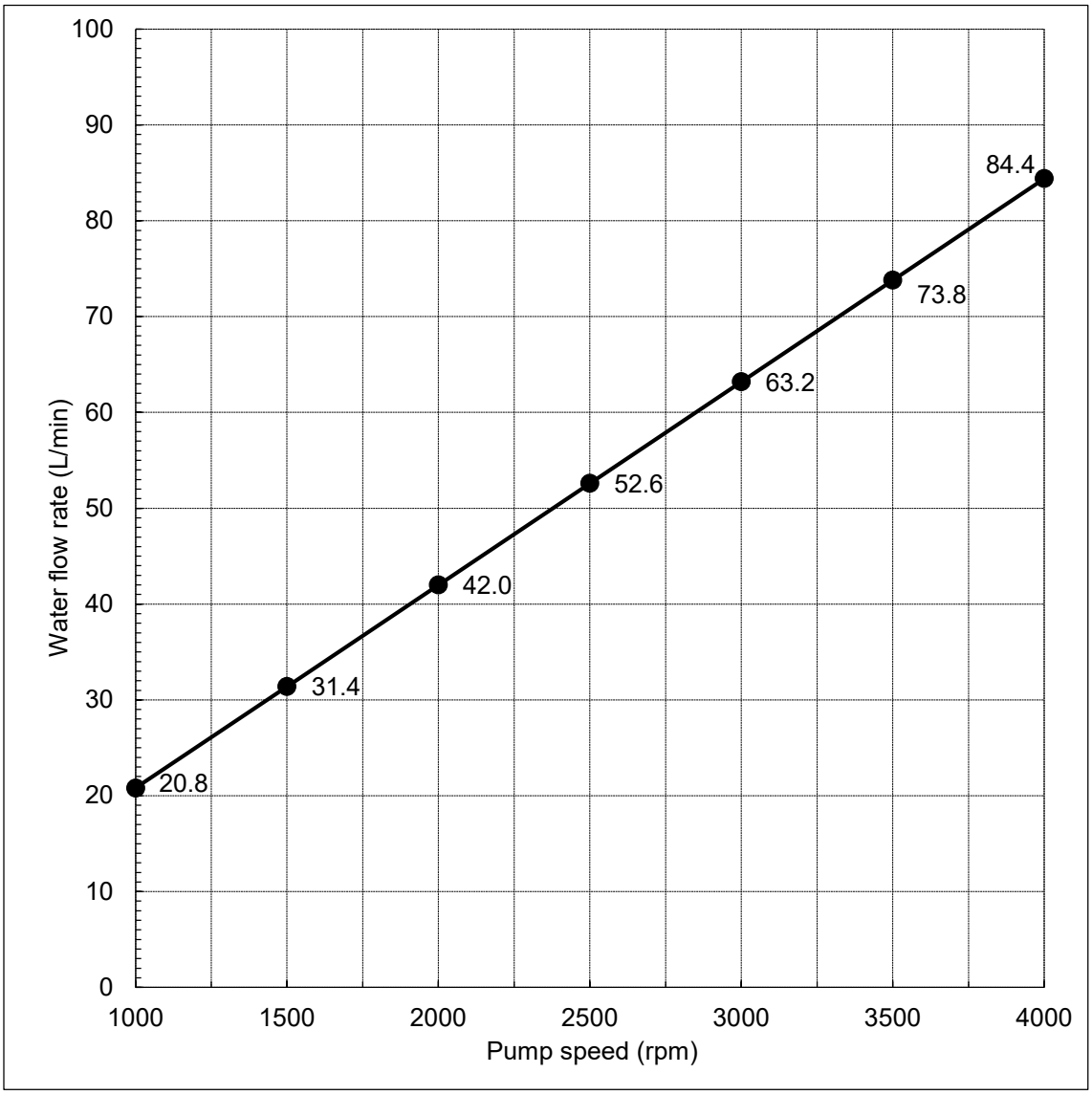


Figure 1

■ **NOTE**

- **The flow is with standard radiator.**
- **Including engine block, radiator and piping resistance.**

[8] CENTER OF GRAVITY

1. with standard flywheel and rear end plate

Model	Dry weight kg (lb)	Center of gravity		
		X mm (in)	Y mm (in)	Z mm (in)
WG1605-GL-E3	122 (269)	-1 (-0.039)	95 (3.740)	182 (7.170)

■ **NOTE**

- **Cooling water and lubricating oil weight is not included in above engine weight.**

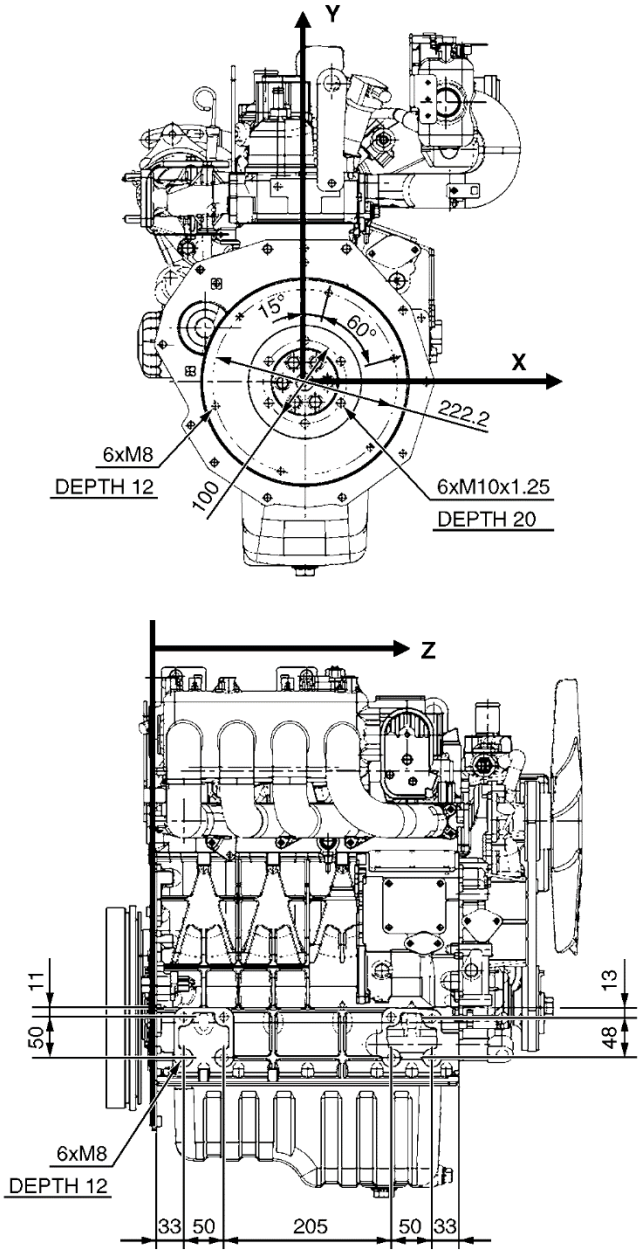


Figure 2

[9] MASS ELASTIC SYSTEMS

(1) Equivalent Torsional Vibration Data

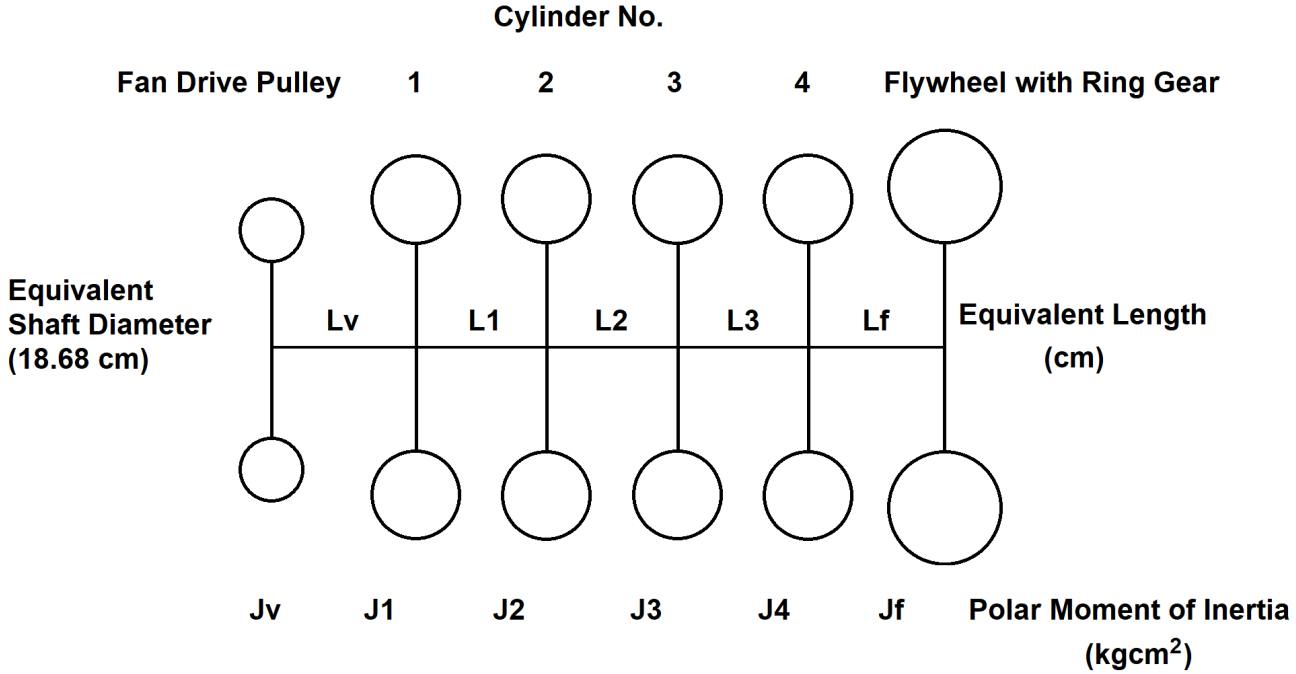


Figure 3

MODEL	EQUIVALENT LENGTH (cm)					POLAR MOMENT OF INERTIA (kgcm^2)					
	Lv	L1	L2	L3	Lf	Jv	J1	J2	J3	J4	Jf
WG1605	20200	2520	2520	2520	1863	12.7	39.2	39.2	39.2	39.2	978

■ **NOTE**

- **Flywheel EG523-2501Δ, Fan Drive Pulley 16249-7428Δ**

[10] UNBALANCED FORCES

(1) Base Data

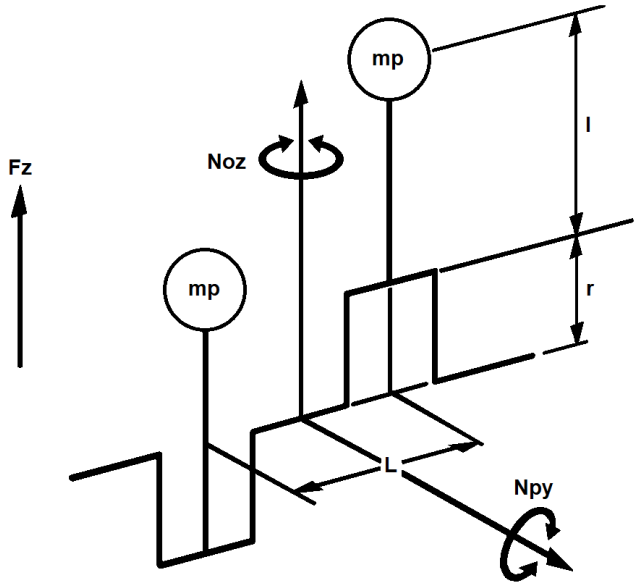


Figure 4

Fz: Unbalanced inertia force
Npy, Noz: Unbalanced inertia couple
mp: Reciprocating mass
r: Crank radius
l: Center distance of connecting rod
L: Cylinder distance
ω: Angular velocity
$2\pi n/60$
n: Engine speed (rpm)

Model	l (m)	r (m)	L (m)	mp (kg)	Bore (mm)	Stroke (mm)
WG1605	0.116	0.0392	0.085	0.69	79.0	78.4

(2) Unbalanced Inertia Force and Couple

(x ω^2)

Model	No. of Cylinder	Cylinder Bore (mm)	Order	Fz	Npy	Noz
WG1605	4	79.0	1	0	0	0
			2	0.02525	0	0

(3) An Example of Calculation

Calculation condition	ω^2	Fz, Npy, Noz		
			Order	Calculation
Engine model: WG1605 Engine speed: 3600 rpm	$(2 \times \pi \times 3600/60)^2 = 142122$	Fz (N)	1	0
			2	0.02525 × 142122 = 3589
		Npy (Nm)	1	0
			2	0
		Noz (Nm)	1	0
			2	0

REVISION HISTORY

File Name	Remarks	Date
KORD3_19-143_ TECHNICAL_INFORMATION_FOR_WG1605.pdf	Typo correction	Jul. 17, 2019
KORD3_20-611_ Technical_information_for_WG1605.pdf	Document style standardization	Dec. 25, 2020